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To
The Recruitment Officer
Huawei

Dear Recruitment Officer,

I am writing this to you wondering if you anticipate any opening research positions in wireless communications (5G NR/Beyond). Currently, I am working with Prof. Lajos Hanzo and Prof. Mohammed El-Hajjar in Next Generation Wireless Group at the University of Southampton in liaison with InterDigital, where I am in the last stages of writing my Ph.D. thesis titled "Intelligence-Aided Transceiver Design for Millimeter Wave Communications". During the course of my Ph.D., I have published a dozen IEEE peer-reviewed papers that include **9 IEEE journals** and **3 IEEE conference papers**, while 2 papers are under review process. My Ph.D. study has been funded by **InterDigital** Europe, where our collaboration resulted in filing **3 international patents**.

My work is mainly focused on designing the transceivers for millimeter wave systems relying on sophisticated machine learning and deep learning techniques. More particularly, I have proposed a radial basis neural network assisted channel prediction to circumvent the problem of channel aging, which is especially more pronounced in massive multiple-input multiple-output. Furthermore, I have developed a deep learning scheme relying on soft-max neural network for beam-alignment in mmWave communications. Additionally, to overcome the pilot-overhead in the channel estimation, I have proposed a semi-blind detection algorithm by invoking neural networks; while the KNN supervised learning is invoked to employ link-adaptation to circumvent the reliance on look-up table based SNR threshold in the face of the hardware impairments and time-varying wireless channel.

Previously, I have demonstrated my skills in wireless communications through a research project at Defence Research and Development Organization, Govt. of India, where I implemented amplitude modulation on a software defined radio (SDR). The challenge was to interface FPGA with DSP processor, and RF section of SDR. Signals were sent through DSP processor, sampled and modulated using FPGA, and fed to transmitter of the RF section. The transmitted signals were received and played on a speaker with good quality output and reproduced well. Additionally, I have got an opportunity to work with Alcatel Lucent during the summer of 2013, where I worked on Gigabit Passive Optical Networks. My particular work involved developing ONT Management Control Interface, a protocol which acts an interface between Optical Networking Terminal and Optical Line Terminal.

While pursuing my Ph.D., I have also co-supervised 3 undergraduates students in their dissertation, where my work involved in teaching and helping the students understand the advanced

wireless transmission techniques, ensued by brain-storming sessions. These sessions came to a fruition in the form of 1 IEEE peer-reviewed conference and 1 IET peer-reviewed journal.

With these words I present my application; and I look forward to hearing from you to discuss the position further.

Sincerely,
K. Satyanarayana.